Reel Rocket -Product Requirements Document Template

> 💡 **TemplateInstructions**: This template follows industry best practices from major tech companies. Replace bracketed placeholders with your specific content. Each section includes guidance on what to include.

📋 DocumentInformation

Field	Value
ProductName	Reel Rocket- Ai powered – Reel Coach for small businesses
DocumentOwner	Aniesh Das
Stakeholders	Product Manager, Engineering Lead, Design Lead, Data Science Lead, App Dev
LastUpdated	06/07/25
Status	Draft/ In Development
Version	1
DocumentType	Platform

🎯 ExecutiveSummary

Problem: Small business owners lack the confidence and tools to create effective Instagram Reels, Low Reels adoption due to skill, trend fatigue, and lack of editing know-how.

Solution: AI-assisted scriptwriting, audio suggestions, visual auto-editing, and scheduling in regional languages.

Impact: Boosts weekly Reels adoption, branding consistency, and engagement.

Investment: High ROI due to increased visibility and organic growth—Reels provide 2x more reach than regular posts.

📊 ProblemStatement

CurrentStateAnalysis

Most small business owners rely on ad-hoc help from relatives or freelancers and rarely post Reels.

KeyPainPoints:

- Low confidence in video creation
- Lack of trend awareness
- No time for editing and posting
- Branding inconsistencies

RootCauseAnalysis

Small business owners in India — from boutique clothing sellers to local bakery owners — understand that **Instagram Reels drive visibility**.

➡️ **Why this matters**:
Instagram Reels offer an organic way to reach a wider audience without paid ads.

Pain Points Identified:

- **They lack time, skill, or confidence** to script, shoot, edit, and publish quality Reels regularly.
  🟨 *Barrier to entry*: Even if they know Reels work, execution is intimidating and time-consuming.
- **They struggle to keep up with trends** (songs, templates, transitions).
  🔁 *Trend fatigue*: Constant updates in Reels culture overwhelm non-digital natives.
- **Many rely on young relatives or freelancers**, leading to inconsistent branding and delays.
  🧩 *Dependence problem*: External help creates dependency and reduces control over brand identity.
- **Reels are either too generic or too complex** to execute solo.
  ⚖️ *Execution paradox*: Tools like Canva exist, but without guidance, results are either boring or unusable.

CostofInaction

📉 **Lost Visibility & Organic Growth**

- Instagram Reels have **2x higher reach** than regular posts.
  → By not using Reels, businesses **miss out on 50–60% of potential organic exposure**.

💸 **Reduced Customer Acquisition**

- Without regular Reels, discovery through Instagram's algorithm drops.
  → This leads to **fewer new followers and walk-in customers**, especially for hyperlocal businesses.

🕒 **Increased Time & Cost Dependencies**

- Relying on freelancers or family leads to **delays, inconsistent quality**, and **higher recurring costs**.

📉 **Stagnant Brand Identity**

- Without visual storytelling, brands struggle to **differentiate themselves**.
  → This results in **low recall and trust**, especially in saturated local markets.

📊 **Competitive Disadvantage**

- Competitors using Reels consistently will rank higher in feeds and hashtags.
  → Non-adopting businesses risk becoming **invisible or irrelevant** in digital discovery.

📉 **Missed Revenue Opportunities**

- Seasonal campaigns, product launches, or offers go unnoticed.
  → Could lead to **10–20% drop in potential monthly sales**, especially during festivals or promotions.

➡️ *Opportunity framing*: Highlights a wide, underserved market with strong intent but poor enablement.

🎯 Goals and Success Metrics

Primary Objective

We want to automate the tedious process of making simple reels and provide a wide range of templates to our users and help them to become social media marketing savvy in minutes!

Success Metrics

🥇 Level 0 Metrics (Primary KPIs)

- Weekly Active Users (WAUs)
- Number of Reels published via app

🥈 Level 1 Metrics (Secondary KPIs)

- Avg. editing time saved
- % Reels using suggested templates/audio

🥉 Level 2 Metrics (Engagement/Quality)

- Completion rate of Reels
- Engagement per Reel (likes/comments)
- Feedback score

🛡️ Guardrail Metrics

[Metrics that should not degrade while pursuing primary goals]

- User churn rate
- Number of support tickets

👥 Target Audience Primary Users [Primary User Segment]

- **Size:** Around 5 Miilion Small business owners with low tech knowledge
- **Characteristics**: Over the age of 40, less tech savvy
- **Current Behavior**: Staying away from social media and digital marketing or hiring others and outsourcing
- **Motivation**: Spending money on somethinng relaviively cheap and free with high visibility

UserPersonas

Persona 1: Priya, 29 — Boutique Owner in Jaipur

- Tech Comfort: Moderate (uses Instagram, Canva)
- Goals: Increase local visibility, boost festive sales
- Pain Points: Editing Reels, following trends, voiceover recording
- Trust Needs: Simplicity, regional language support

Persona 2: Salim, 42 — Bakery Owner in Hyderabad

- Tech Comfort: Low (uses WhatsApp, YouTube occasionally)
- Goals: Attract walk-ins, improve brand recognition
- Pain Points: No time or skill to edit content, unsure when to post
- Trust Needs: AI guidance, hands-free assistance

💡 SolutionOverview

High-LevelApproach

ReelRocket is designed as a plug-and-play AI co-pilot that simplifies Reel creation for small business owners through the following approach:

1. **Input → Intelligence → Output**
    - Users input basic product/business details.
    - AI generates scripts, suggests templates, and selects matching audio.
    - The system auto-edits the final Reel using uploaded images and outputs a ready-to-post video.
2. **No-Edit, No-Stress Workflow**
    - Eliminate manual editing by leveraging auto-sync features (voiceover, transitions).
    - Optimize for mobile usability with step-by-step guidance.
3. **Personalization + Trend Alignment**
    - Content templates are tailored to industry type and language preference.
    - Audio and visuals align with current Instagram Reel trends.
4. **Smart Scheduling**
    - The tool recommends best times to post based on engagement data and offers 1-tap scheduling

KeyPrinciples

✅ **Simplicity First**
Design for non-technical users with zero video editing experience. Every step must be intuitive and minimal.
✅ **AI-Assisted, Not AI-Replaced**
The AI suggests and automates, but the user retains control over the final content to preserve brand voice.
✅ **Hyper-Localization**
Support for regional languages, festival themes, and local cultural relevance to increase relatability.
✅ **Speed + Consistency**
Reduce the time to publish from hours to minutes, while maintaining visual and audio branding.
✅ **Trend Responsiveness**
Continuously update templates and audio suggestions to reflect ongoing Reel trends.
✅ **Mobile-First Design**
Ensure the full workflow (script → edit → publish) works seamlessly on mobile for business owners on the go.
✅ **Trust Through Transparency**
Offer explainable AI suggestions and editable templates so users always understand and control their output.

SolutionComponents

🔩 DetailedRequirements

✅ **Feature 1: AI Script Generator**

- **Objective:** Help users auto-generate short, trend-based scripts
- **User Story:** As Priya, I want script prompts so I can create content fast
- **Functional Requirements**: GPT API integration, template tagging by industry
- **Backend/Frontend**: Use GPT API, script templates
- **Acceptance Criteria:** Given a user has entered their product/business info,
  When they click "Generate Script", Then system returns editable script in < 3 seconds using relevant templates.
- **Priority:** Medium
- **Effort Estimate:** 13

✅ **Feature 2: Audio + Template Suggestions**

- **Objective:** Suggest trending audio and matching Reel templates
- **User Story:** As Salim, I want music ideas so my videos feel current
- **Functional Requirements:** Integrate Instagram trending audio API or scraper
- **Backend/Frontend:** Audio database, category filtering
- **Acceptance Criteria**: Given a user has selected their category or script,
  When they reach the "Audio + Template" screen, Then the system should display at least 3 relevant trending audio-template pairs within 3 seconds.
- **Priority:** Medium
- **Effort Estimate:** 5

✅ **Feature 3: Auto Reel Maker**

- **Objective**: Convert images + text to Reels with voiceover and effects
- **User Story:** As Priya, I want my product photos to turn into professional videos
- **Functional Requirements:** Use FFmpeg, TTS, and image transition libraries
- **Backend/Frontend:** Photo input → FFmpeg video render pipeline
- **Acceptance Criteria**: Given a user has uploaded 3+ images and selected a script/audio,
  When they tap "Create Reel", Then the app should generate a preview video with synced transitions and voiceover, with >90% accuracy in text/audio alignment.
- **Priority:** High
- **Effort Estimate:** 13

🎨 User Experience

User Journey Map:

Stage	Action	Emotion
1	Enters product info	Confused → Curious
2	Script & Audio Suggested	Curious → Hopeful
3	Uploads images	Hopeful
4	Auto Reel preview	Excited
5	Publish/Schedule	Satisfied → Proud

Key Interaction Points

User experiences satisfaction and relief once they find out the reel is completed and uploaded

Design Principles

1. User-Centric Simplicity
- **Why**: Most users are non-technical and short on time.
- **How**: Reduce friction by limiting steps, using familiar interfaces (e.g., WhatsApp-style layout), and clear CTAs like "Generate Script" or "Create Reel".

2. AI as a Co-Pilot, Not a Replacement
- **Why**: Users want control and trust in automation.
- **How**: AI provides suggestions, not commands — users can review, tweak, and personalize the output.

3. Mobile-First, Lightweight Experience
- **Why**: Most small business owners operate from mobile devices.
- **How**: Prioritize vertical video previews, tap-to-select templates, voice inputs, and offline drafts.

4. Language Inclusivity
- **Why**: India's linguistic diversity demands localized support.
- **How**: UI and scripts are available in English and key regional languages (e.g., Hindi, Tamil, Telugu).

5. Trend Responsiveness
- **Why**: Reel trends evolve daily.
- **How**: Built-in trend tracker for songs, filters, and templates to keep users updated automatically.

6. Speed & Efficiency
- **Why**: Users are busy solopreneurs.
- **How**: End-to-end Reel generation (script + visuals + audio + scheduling) in <5 minutes.

7. Consistent Branding Support
- **Why**: Inconsistent content weakens brand identity.
- **How**: Save logos, fonts, color themes and auto-apply them across Reels.

8. Guided Confidence
- **Why**: Many users lack confidence in content creation.
- **How**: Step-by-step tooltips, voice guidance, and default templates help first-timers feel supported.

9. Trust through Transparency
- **Why**: AI black boxes reduce trust.
- **How**: Explain why a certain script, audio, or post time was suggested ("Based on bakery trends in Hyderabad...").

10. Accessibility
- **Why**: The product must work for a wide range of users with different needs.
- **How**: Support voice navigation, large text display, and simple layouts with no hidden interactions.

🛠 Technical Requirements

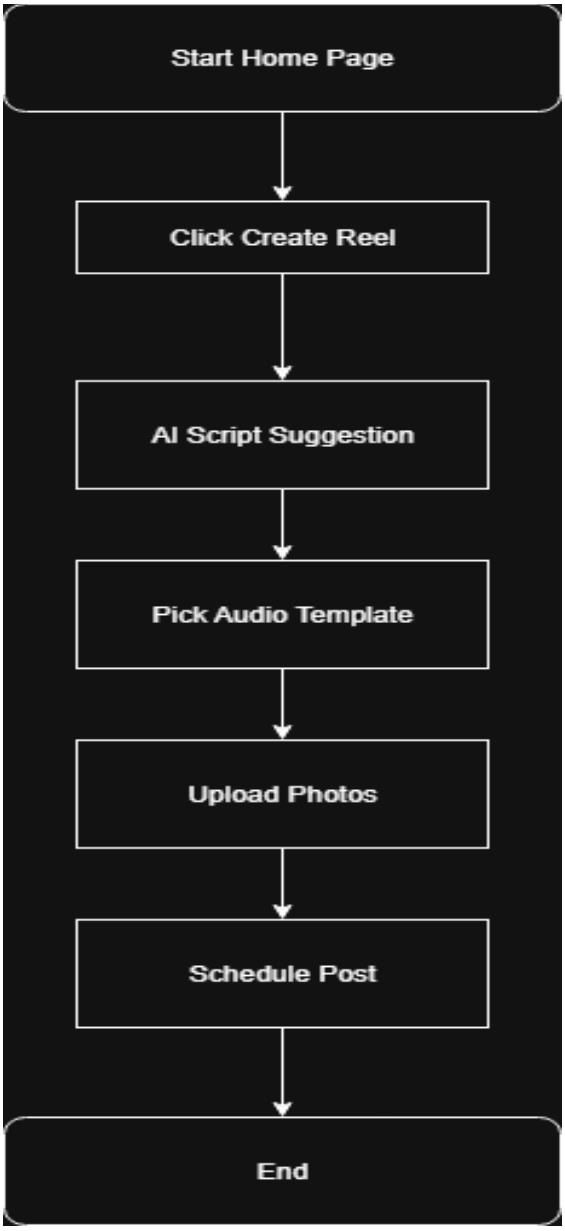
Architecture Overview

- **Frontend**: React (mobile-friendly), audio & image upload, previewer
- **Backend**: Node.js/Python Flask, MongoDB for storage
- **APIs**:
  - GPT (script generation)
  - FFmpeg (video creation)
  - Instagram Graph API (posting/scheduling)

Wireframes

Create screens for:

- Home/Dashboard
- Script Suggestions
- Audio + Template Picker
- Reel Preview & Publish Screen



Performance Requirements

✅ Response Time

- **Script Generation**: < 2 seconds (after product info is submitted)
- **Template & Audio Suggestions**: < 3 seconds
- **Reel Preview Rendering**: < 5 seconds (for image-based input + voiceover)
- **Page Load Time**: < 1.5 seconds on mobile networks (4G)

🚀 Scalability

- Must support:
  - **Initial load**: 5,000 MAUs (monthly active users) at launch
  - **Growth scale**: Scalable to 100,000+ MAUs within 12–18 months
- **Backend design**: Use microservices and caching for faster load distribution
- **Media Processing**: Cloud-based or edge-rendering (FFmpeg or similar) to handle burst loads during festival seasons or promotions

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🟢 Availability

- **Uptime Requirement**: 99.5% monthly uptime (target downtime < 3.6 hours/month)
- **Failover Strategy**: Auto-retry for failed API calls (especially for GPT, Instagram API)
- **Backup**: Daily snapshot of user-generated content (scripts, Reels, drafts)

🔐 Security Requirements

- **User Authentication**: OAuth 2.0 / OTP-based login for easy and secure access
- **Data Privacy**:
  - Encrypt user content (scripts, uploaded images, voice data) at rest and in transit (TLS 1.2+)
  - Anonymize any usage data collected for trend analysis
- **Access Control**:
  - Role-based access for admin, moderator, and user (future-proofing for scale)
- **Storage Security**:
  - Secure S3/Blob storage with private access tokens for all media assets
- **Third-Party APIs**:
  - Compliant with Instagram Graph API policies (for scheduling/posting)
  - Rate-limiting and token refresh logic to prevent abuse

📅 ImplementationPlan

Phase1:MVP(MinimumViableProduct)
Timeline: 1st Week
Scope: App Layout
SuccessCriteria: succesful working Hi Fidelity Wireframe
Phase2:[EnhancementPhase]
Timeline: 2nd Week
Scope: Additional features or improvements
SuccessCriteria: Final version of app
Phase3:[FutureIterations]
Timeline: 3rd Week
Scope: Fully working app

⚠️ Risks and Mitigation

High-Risk Items

Technical Risks

1. Risk: AI-generated scripts/audio may not align with user's brand tone

- **Impact**: Medium
- **Probability**: High
- **Mitigation**: Allow users to choose tone presets (e.g., quirky, formal, friendly) and provide edit options. Use feedback loops to improve relevance over time.

2. Risk: Reel rendering may fail or lag on low-end mobile devices

- **Impact**: High
- **Probability**: Medium
- **Mitigation**: Optimize for mobile-first performance using compressed assets, offer preview mode in low-res, and offload rendering to cloud-based workers.

3. Risk: Instagram API limitations for posting or scheduling

- **Impact**: Medium
- **Probability**: Medium
- **Mitigation**: Provide fallback option to download and manually post Reels. Monitor API policies and maintain compliance. Buffer tokens and refresh logic.

📈 Business Risks

1. Risk: Low adoption from target users due to tech aversion or confidence gap

- **Impact**: High
- **Probability**: High
- **Mitigation**: Design for simplicity with onboarding tutorials, regional language support, and pre-filled templates. Offer "1-tap reel creation" to lower barrier.

2. Risk: Competitors like Canva, InShot, or Meta's own tools replicate features

- **Impact**: Medium
- **Probability**: Medium
- **Mitigation**: Focus on hyper-localization, AI co-pilot features, and small biz personalization—areas big players ignore. Build community-driven templates.

3. Risk: User drop-off after initial trial due to lack of habit or ROI clarity

- **Impact**: High
- **Probability**: Medium
- **Mitigation**: Add content calendar planner, weekly posting reminders, milestone rewards, and auto-publish to build consistency.

🛠️ Operational Risks

1. Risk: Difficulty maintaining template freshness and trend alignment

- **Impact**: Medium
- **Probability**: High
- **Mitigation**: Automate trend discovery using open-source scrapers or partner APIs; set weekly update cycles with content curators.

2. Risk: High support demand from first-time users unfamiliar with AI

- **Impact**: Medium
- **Probability**: Medium
- **Mitigation**: Provide contextual tooltips, chat-based help, and voice-based guidance. Build a simple FAQ and WhatsApp support flow.

3. Risk: Scaling infrastructure cost as users grow and media usage increases

- **Impact**: High
- **Probability**: Medium
- **Mitigation**: Use serverless architecture and cloud auto-scaling for rendering workloads. Monitor usage metrics and cache static components.

🔗 Dependencies

🔧 Internal Dependencies

✅ **Team/System: AI & Backend Engineering Team**

- **What is needed**: GPT-based script generation and template recommendation logic
- **By when**: MVP prototype completion by **Week 3** of development sprint

✅ **Team/System: Media Rendering Module (Video + Voiceover)**

- **What is needed**: Integration of FFmpeg (or equivalent) pipeline for image-to-Reel conversion, synced with AI voice
- **By when**: Working pipeline ready by **Week 4**

🌐 External Dependencies

✅ **Third Party/Partner: OpenAI or Ollama (for language model)**

- **What is needed**: Access to GPT API or local LLM model for generating scripts and trend-based copy
- **By when**: Set up credentials and basic prompt tuning by **Week 2**

✅ **Third Party/Partner: Instagram Graph API**

- **What is needed**: Access and approval for Reels scheduling and posting capabilities
- **By when**: API integration and token management configured by **Week 4**

⛔ Blocking Dependencies

- **1. Instagram API Access Approval**
  - Without this, **auto-posting and scheduling** features cannot be implemented or tested.
- **2. AI Model Selection & Prompt Architecture Finalization**
  - Must finalize the model (GPT/Open-source) and prompt design before **any script or template generation** work begins.
- **3. FFmpeg / Media Rendering Stack Decision**
  - Video rendering tech stack (cloud-based or local) must be finalized early to inform infrastructure setup.

📈 Analytics and Measurement

Key Events:

1. Script generated
2. Audio/template selected
3. Reel previewed
4. Reel published
5. Reel scheduled
6. Feedback submitted
7. Reel edited
8. User retention day 7

AARRR Metrics:

- **Acquisition**: Signup source (referral, web, ad)
- **Activation**: First successful Reel
- **Retention**: Reels created per week
- **Referral**: Users who invite others
- **Revenue**: Optional freemium features

A/B Test:

Test "auto-scheduling" vs "manual post" for Reel engagement.

PR-FAQ

Press Release:

Introducing ReelRocket: Your AI Coach for Instagram Reels
ReelRocket is an AI assistant built for India's small businesses—helping you script, create, and post high-quality Reels without the hassle. Whether you're a bakery in Hyderabad or a boutique in Jaipur, ReelRocket helps you shine on Instagram.

FAQs:

1. **Why did we build this?**
   To help time-starved business owners grow using Reels.
2. **How does it work?**
   Input product info → Get script/audio → Upload photos → Preview Reel → Publish.
3. **What's next?**
   Upcoming features: multi-lingual voiceover and community template sharing.
4. **How will success be measured?**
   User engagement, Reel output per week, and feedback ratings.
5. **Is it free or paid?**
   Free basic tier. Premium features like auto-posting and analytics are paid.
6. **How is it different from Canva or InShot?**
   Unlike Canva/InShot, it's AI-guided, trend-based, and designed for business owners with no editing experience.

🚀 **Go-to-Market Strategy**

📢 **Launch Strategy**
🎯 **Target Segment**

- Small business owners in India (boutiques, bakeries, salons, gift shops, etc.)
- Primarily mobile users with low-to-medium tech literacy
- Prioritize Tier 1 and Tier 2 cities with high Instagram usage

1️⃣ **Phased Rollout**
✅ **Phase 1: Private Beta (Weeks 1–3 post-dev)**

- Hand-pick 50 users from waitlist (focus: boutique owners and food sellers)
- Provide 1:1 onboarding support via WhatsApp
- Collect feedback on usability, script relevance, and rendering experience

✅ **Phase 2: Early Access Launch (Week 4)**

- Open to broader waitlist of 500–1,000 users
- Launch simple landing page with signup + walkthrough demo
- Incentivize referrals: "Create 3 Reels, invite 2 friends — unlock premium features"

✅ **Phase 3: Public Launch (Week 6)**

- Social media campaign showcasing Reels made with ReelRocket (Reels of Reels)
- Partner with Instagram content coaches and micro-influencers
- Feature on startup/creator platforms (e.g., Product Hunt, YourStory, LinkedIn post)

📣 **Marketing Channels**

- **WhatsApp**: Broadcast messages with tutorials, quick tips, and reminder nudges
- **Instagram**: Carousel ads, Reels showing how ReelRocket works, user-generated Reels
- **YouTube Shorts:** 15-sec demos of "Before vs After" with ReelRocket
- **Regional Influencers:** Partner with vernacular creators for credibility and relatability

🎁 **Promotional Offers**

- **Free plan:** Access to limited scripts/templates
- **Premium plan:** Unlock auto-posting, brand kits, and advanced transitions

📣 Communication Plan

🔷 Internal Communication

- Weekly standups (Slack + Notion updates) for design, engineering, and PM teams
- Sprint demos every 2 weeks to leadership and stakeholders
- Jira or Trello board for feature progress tracking and dependencies
- Email digest for senior leadership updates every fortnight

🔷 External Communication

- Early access waitlist launch via website + WhatsApp
- Launch email with how-to video tutorial
- In-app onboarding guide + tooltips
- Social media campaign showcasing success stories and Reels made with ReelRocket

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🧠 Training Requirements

- **Support Team**:
  - Product walkthrough with FAQs and common troubleshooting
  - Escalation flow for AI issues (e.g., bad script suggestions)
- **Sales/Outreach Team**:
  - Pitch deck and demo videos
  - Feature-benefit mapping by user persona
  - Pricing tiers and differentiators from competitors (Canva, InShot)
- **Customer Success/Community**:
  - Onboarding flow understanding
  - Community engagement playbook (template sharing, feedback collection)

🚀 Launch Criteria

- MVP features fully functional:
  - Script Generator
  - Template + Audio Suggestion
  - Reel Auto-Creation
  - Download or Instagram Scheduling
- All critical bugs fixed
- Tested on 5+ real user devices (low-mid-high range phones)
- Positive feedback in user pilot (>80% said they would use it again)
- Support content + training in place
- Landing page live with onboarding funnel

🎯 Success Criteria

✅ Definition of Success

- 1,000+ Reels created in first 30 days
- 60% users complete a Reel from start to publish
- 4.5⭐ average user feedback
- <5% drop-off at each core step (script → edit → publish)

📋 Launch Readiness Checklist

- ✅ All acceptance criteria met
- ✅ Performance benchmarks achieved (render time, response time)
- ✅ Security review completed (storage, user privacy)
- ✅ Analytics tracking implemented (AARRR events)
- ✅ Documentation completed (user + developer)
- ✅ Training delivered to internal teams
- ✅ Rollback plan prepared in case of system failure or user issues

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🔍 Post-Launch Review

- **Timeline**: 14 days and 30 days post-launch
- **Review Process**:
  - Analyze engagement metrics (Reels created, features used, DAU/WAU)
  - Collect NPS + qualitative feedback
  - Hold retrospective to assess gaps in product, onboarding, and support
  - Prioritize v1.1 roadmap features based on findings

📚 Appendices

Appendix A: Competitive Analysis

- **Canva**: Great for design, lacks AI and trend focus
- **InShot**: Video editor, but manual and not tailored to small businesses
- **Instagram Reels**: Native but lacks guidance or scripting
- **ReelRocket Advantage**: AI-led, one-tap creation, localized, small biz-focused

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Appendix B: User Research

- 10 user interviews (boutique owners, bakers, stylists)
- 90% cited "lack of time + knowledge" as barrier to using Reels
- 7/10 said they rely on family/friends to help post
- All wanted a tool that tells them *what to say and how to say it*

Appendix C: Technical Deep Dive

- Tech stack: React (frontend), Python/Node (backend), FFmpeg (render), GPT or Ollama (LLM), Coqui TTS
- APIs: Instagram Graph API, OpenAI/Ollama, Trend scraper (custom)
- Cloud storage + queuing architecture for async Reel rendering
- Scalability via microservices and caching

Appendix D: Legal and Compliance

- User data stored securely and encrypted
- Content ownership: User retains rights over uploaded images and generated Reels
- T&C updated for AI-generated content attribution
- API usage compliant with Meta and OpenAI policies

✅ Approval

Role	Name	Signature	Date
Product Manager	[Name]		
Engineering Lead	[Name]		
Design Lead	[Name]		
Data Science Lead	[Name]		
Business Stakeholder	[Name]		

📝 ChangeLog

Version	Date	Changes	Author
1.0	[Date]	Initial draft	[Name]
1.1	[Date]	[Description of changes]	[Name]